MVP Validation Findings & Refactoring Priorities Form

Section 1 — Project Information

1. **Team/Group Name:** 2627-sem1-it411-53
2. **Project/System Title:** Elementopia 2.0
3. **Program / Section:** BSIT - IT411
4. **Team Members:**
    a. Danielle Maxine P. Ocampo
    b. Clyde C. Benolirao
    c. Erica Y. Dabalos
    d. Wayne Kenji B. Dibdib
    e. Jian Marc Ceniza
5. **Adviser:** Dr. Patrick L. Bacalso
6. **System URL / MVP Link:** https://elementopia-frontend.vercel.app/
7. **Date of MVP Validation:**
    a. Initial MVP Validation: Sept. 19, 2026
    b. Reinforced MVP Validation: Oct. 09, 2026
8. **Number of respondents/participants:**
    a. Initial MVP Validation: 30 Grade 8 Students
    b. Reinforced MVP Validation:
        i. 30 Grade 8 Students: 20 responded
        ii. 30 Grade 9 Students: 15 responded
9. **Types of respondents involved**
    a. End User/s: Grade 8 and Grade 9 Students (Total = 65 Students)
    b. Subject Matter Expert: High School Science Teachers (4 Teachers)
    c. Decision Maker: Capstone Adviser and academic evaluators assessing curriculum alignment.

Section 2 — MVP Validation Overview

1. **What was the primary purpose of your MVP validation?**

   The primary purpose of our MVP validation was to empirically determine whether Elementopia's core gameplay mechanics—specifically the interactive drag-and-drop synthesis workbench, contextual 'Meaningful Byproduct' error alerts, the Hazmat Protocol failsafe (sliding-window anti-guessing lock), and real-time multiplayer challenges—effectively stimulate student engagement (evaluated via Keller's ARCS Motivational Model) while producing verifiable cognitive learning gains in DepEd Grade 8 and Grade 9 Chemical Bonding (evaluated via Bloom's Taxonomy and Hake's normalized gain (g)). Concurrently, the validation engaged licensed science educators under the TPACK and UPA frameworks to assess curriculum fidelity, pedagogical scaffolding across stages, and practical classroom viability, ensuring the software transitions from an entertainment prototype into an academically sound educational tool.

2. **What validation framework/model did your team use?**
   - **Framework/Model: Multi-tiered Evaluation Model combining:**
     - Keller's ARCS Model of Motivational Design (Attention, Relevance, Confidence, Satisfaction) for student motivational engagement;
     - Bloom's Revised Taxonomy / Hake's Normalized Learning Gain (g) for objective cognitive mastery; and
     - TPACK (Technological Pedagogical Content Knowledge) & UPA (Usability, Pedagogy, Accessibility) for expert Teacher/SME validation.
   - **Key constructs/criteria evaluated:**
     - Student Constructs: Attention (visual curiosity and drag-and-drop mechanics), Relevance (curriculum connection to chemical bonding), Confidence (error recovery after failed attempts), Satisfaction (overall enjoyment and challenge), and Post-Test Mastery Accuracy (% correct on bond types, stoichiometry, and byproduct recovery).
     - Teacher/SME Constructs: Curriculum Alignment with DepEd competencies, Scientific Accuracy of chemical formulas, Pedagogical Scaffolding across room stages, Intervention Efficacy of byproduct feedback, and Classroom Viability.
   - **Why was this framework/model appropriate for your project?**

     Single-dimension usability heuristics fail to prove whether an educational game actually fosters conceptual learning or merely entertains. Triangulating student psychological motivation (ARCS), measurable cognitive learning gains (pre/post diagnostic gain), and expert pedagogical assessment (TPACK) directly addressed the faculty panel's critical question regarding whether our MVP was 'faulty or fully optimized.'
3. **Briefly describe how the validation was conducted.**

   A. Who participated?

   a. 65 secondary students (30 in Phase 1: Initial Baseline, 35 in Phase 2: Reinforced)

b. 4 licensed science teachers.

B. How do they interact with the MVP?

Participants accessed the live web application on school desktop computers, laptops, and mobile smartphones/tablets using modern browsers (Chrome, Firefox, Safari) via frictionless guest onboarding (temporary session nickname).

C. What activities/tasks they performed?

a. Listened to an interactive demo by the developers/researchers.
b. Completed an initial 3-item in-game cognitive pre-test diagnostic;
c. Navigated Elemental Resonance puzzle rooms across a 3-stage progression curve (Stage 1 to 3 of Domain 1);
d. Triggered and observed Meaningful Byproduct alerts and Hazmat Protocol graying upon invalid combinations;
e. Reviewed intermediate stage debriefings (StageTransitionModal) and final domain clearance recaps (DebriefingModal);
f. Participated in real-time competitive multiplayer matches via 5-digit room codes.

D. How feedback/data was collected

The data was observed and collected through automated database telemetry logging (completion speed, accuracy, timestamps). The students then completed a standardized 20-item ARCS Likert questionnaire with post-test section, and open-ended qualitative questions. Additionally, the Teachers completed a SME evaluation rubrics, and open-ended qualitative interviews

Section 3 — MVP Validation Findings

Instruction: Identify the most significant findings from your MVP validation. Do not simply list bugs, cosmetic changes, or minor UI/UX comments. Focus on findings that have implications for the system's ability to achieve its SMART objectives and measurable outcomes.

Finding #1

1. **What did you learn from the stakeholders?**

While students successfully combined elements to clear cavern barriers and expressed high overall satisfaction (ARCS composite = 4.14/5.00), post-test diagnostic evaluation revealed a significant cognitive drop in explicitly identifying chemical bond types. Specifically, while 91.4% (32/35) mastered byproduct recovery, only 68.6% (24/35) correctly classified H2O as a covalent bond. Both students and teachers noted that although the system had end-of-stage summary screens, they functioned primarily as game scorecards (showing time and compound counts) rather than reinforcing chemical bonding theory (electron sharing vs. electron transfer).

2. **What evidence supports this finding?**

Post-test recorded the lowest in the entire evaluation (68.6%). Student interview feedback: 'I knew hydrogen and oxygen make water, but the game didn't tell me why they share electrons instead of giving them away.' Furthermore

a teacher observed: 'The transition between clearing a stage barrier and starting the next stage is too brief. Students need direct reinforcement of the bond type (polar vs. nonpolar covalent) and Lewis structures before moving to harder stages.'

3. **Which stakeholder/user type provided this feedback?**
    - End User: Grade 8 and Grade 9 Students
    - SME: High School Science Teachers
4. **Which SMART objective or measurable outcome is affected?**

    The affected SMART Objective 2, with the goal to demonstrate alignment with the DepEd Science Curriculum (Relevance & Normalized Learning Gain: Achieve a medium-to-high normalized learning gain g >= 0.40 on DepEd chemical bonding competencies). This objective directly hinges on students conceptualizing why compounds bond; the 68.6% score on Q24 demonstrates that without milestone debriefs, the platform underperforms against the 75% curriculum mastery benchmark.

5. **What measurable outcome could potentially be improved?**
    - **Accuracy —** Targets raising post-test chemical bond discrimination accuracy from the observed 68.6% baseline to >= 80%.
    - **Effectiveness —** Elevates normalized learning gain (g) from moderate baseline to high cognitive retention (g >= 0.50) on Bloom's conceptual understanding level.
    - **User Satisfaction —** Closes student dissatisfaction regarding unguided stage transitions, improving the ARCS Relevance score from 4.19 to >= 4.50.
6. **Explain how the finding affects the achievement of the objective. Paragraph response.**

    In the baseline MVP, the synthesis engine successfully confirms stoichiometric combinations, but immediately transitions into an unguided gameplay loop. Consequently, some students rely on procedural trial-and-error (dragging tiles until the barrier opens) without cognitively connecting the mechanical action to chemical bonding principles (valence electron pairing, electronegativity differences, and octet stability), even though tips and notes were provided. Since post-test evaluations measure conceptual discrimination rather than gameplay speed, omitting explicit milestone debriefing directly prevents students from reaching the mandated 75% mastery threshold and restricts the normalized learning gain (g). Implementing structured milestone debriefings at stage barriers and domain completion ensures that students reflect upon and synthesize the core scientific concepts before advancing, directly securing the target cognitive gain.

Finding #2

1. **What did you learn from the stakeholders?**

    The initial MVP architecture recorded student pre-test diagnostic baselines exclusively within browser local storage (localStorage). When students switched devices, refreshed cached sessions, or cleared browser data, their pre-test

baselines became disconnected from their subsequent gameplay telemetry in the Supabase PostgreSQL database.

2. **What evidence supports this finding?**

   During Phase 1 testing, 12 out of 30 student baseline records were fragmented due to browser cache resets, requiring manual paper reconciliation. Database inspection confirmed that while session gameplay logs committed reliably, no centralized cloud table existed in Supabase for pre-test cognitive baselines. Evaluators noted that automated calculation of learning gains (g) was impossible without an unbroken, server-side pairing between pre-test and post-test scores.

3. **Which stakeholder/user type provided this feedback?**
   - End User (Students)
   - Decision Maker (Adviser & Teacher Evaluator Analytics).
4. **Which SMART objective or measurable outcome is affected?**

   SMART Objective 2 (Learning Gain Calculation) and SMART Objective 4 (Satisfaction & Automated Cloud Telemetry Logging): 'Automatically record and log student performance metrics, completion velocity, and error patterns into Supabase cloud persistence within under 2 seconds per session, enabling real-time dashboard analytics.' Mathematically, OBJ-02 requires paired pre-test and post-test scores to compute g = (Post - Pre) / (100 - Pre); isolating pre-tests in local storage breaks this equation.

5. **What measurable outcome could potentially be improved?**
   - **Efficiency —** Eliminates manual data collation and paper test reconciliation, automating cloud-side gain computation instantly upon session close.
   - **Reliability —** Increases pre-test diagnostic cloud capture rate from an unstable 60% (12 of 30 lost in Phase 1 cache resets) to 100% persistent cloud availability.
   - **System Response Time / Performance —** Enforces a sub-2-second asynchronous HTTPS write latency (POST /api/telemetry/pretest) without blocking initial gameplay onboarding.
6. **Explain how the finding affects the achievement of the objective. *Paragraph response.***

   The mathematical integrity of SMART Objective 2 and the automation mandate of SMART Objective 4 depend on a continuous, uninterrupted telemetry pipeline. When pre-test diagnostic records are isolated in browser localStorage, any client-side event—such as switching between school computer lab terminals, clearing browser cookies, or opening private browsing sessions—permanently severs the student's baseline diagnostic from their cloud gameplay records. This fragmentation prevents the system from automatically computing individual learning gains in Supabase and deprives the Mastery Dashboard of its baseline comparator. Centralizing pre-test telemetry via an asynchronous cloud endpoint guarantees 100% data pairing, ensuring that student learning trajectories are reliably tracked and verifiable for academic evaluation.

Finding #3

1. **What did you learn from the stakeholders?**

   As students commence Opponent Challenge matches, if one student experienced a cellular network drop or closed their mobile browser tab, the connected peer remained indefinitely trapped in an active match state with a frozen countdown timer, receiving neither a win declaration nor a graceful exit option.

2. **What evidence supports this finding?**

   Observed in 3 out of 18 multiplayer match trials during Phase 2 testing over mobile 4G connections. Student interview feedback: 'My screen stayed stuck on the countdown when my classmate's data disconnected, and I had to refresh and lost my score.'

3. **Which stakeholder/user type provided this feedback?**
   - End User (Competitive Multiplayer Students).
4. **Which SMART objective or measurable outcome is affected?**

   The affected objective was SMART Objective 4 (Satisfaction & Speed-to-Compound Mechanics, *"Achieve a minimum average satisfaction score of 4.2 out of 5.0 on post-interaction surveys evaluating the Mastery Dashboard and Speed-to-Compound mechanics..."* Indefinitely freezing an active player upon peer disconnection severely damages player satisfaction and perceived game fairness, directly threatening the satisfaction threshold for multiplayer.

5. **What measurable outcome could potentially be improved?**
   - **Reliability —** Achieves 100% graceful match resolution by eliminating unhandled WebSocket disconnections and stranded game states.
   - **Usability —** Provides transparent visual feedback (15-second reconnection countdown modal) so the active player understands why the match timer paused.
   - **User Satisfaction —** Protects competitive integrity and prevents player frustration from lost match progress, raising multiplayer satisfaction scores above 4.50/5.00.
6. **Explain how the finding affects the achievement of the objective. *Paragraph response.***

   Achieving SMART Objective 4 requires delivering a dependable, engaging user experience across both the Mastery Dashboard and the competitive Speed-to-Compound module. In the baseline MVP, the absence of an active heartbeat presence monitor means that when a student's cellular data drops or their mobile browser is closed, the remote WebSocket connection silently terminates without notifying the client. This leaves the connected opponent stranded with an indefinite countdown freeze, unable to claim a legitimate win or exit without manually refreshing and losing their own session stats. During validation, this defect resulted in acute player frustration, dragging competitive multiplayer satisfaction down to 3.82/5.0 and directly threatening the mandated 4.20/5.00 benchmark. Introducing a 3-second heartbeat ping with an 8-second silence detector and a 15-second forfeit grace countdown resolves peer dropouts

programmatically, protecting competitive fairness, eliminating stranded match sessions, and directly restoring user satisfaction to meet and exceed the target objective.

Section 4 — Potential Refactoring Priorities

This is the most important section of the form. For each significant finding, identify what should potentially be changed in the system's requirements and/or design.

Refactoring Priority #1: Post-Stage & Post-Domain Educational Debrief

1. **Validation Finding**

   Referencing the Finding #1 from Section 3 (Post-Synthesis Cognitive Gap & Lack of Explicit Bond Discrimination).

2. **What needs to be improved or changed?**

   Formalize and upgrade the prototype's milestone modals into a structured, curriculum-aligned educational debriefing system: StageTransitionModal.jsx for intermediate stage barrier clearance (displaying unlocked masteries and upcoming blueprints) and DebriefingModal.jsx for domain completion (displaying all 12 compounds, accuracy %, and core bonding concepts).

3. **What type of refactoring is required?**
   - **Functional requirement —** Introduces formal software capabilities (FR-1.6) specifying automated modal rendering triggers, mastery checklist generation, and bonding concept explanations.
   - **System process/workflow —** Restructures the stage transition lifecycle from an instantaneous barrier clear into a staged educational reflection checkpoint prior to loading new element sets.
4. **Current MVP approach:**

   The MVP displayed rudimentary summary screens that focused on game metrics (elapsed time, accuracy %, and compound counts) without reinforcing chemical bonding mechanics (electron sharing vs. transfer).

5. **Proposed refactored approach**

   At Stage 1 & 2 Barrier Clearance: Render StageTransitionModal displaying Doctor Atom debriefing, a checklist of unlocked masteries (e.g., binary sharing, octet balance), and a blueprint preview of the upcoming compound. At Stage 3 Domain Clearance: Render DebriefingModal presenting overall accuracy %, time elapsed, all 12 synthesized compound badges with formulas, and a formal explanation of the core bonding concept mastered.

6. **Why is this change necessary?**

   The validation data revealed that mechanical success in synthesizing compounds does not inherently translate into conceptual understanding of

chemical bonding. While 91.4% of students successfully cleared cavern barriers, only 68.6% correctly classified bond types on post-test Question 24. Without structured milestone debriefing, students treat synthesis as an isolated trial-and-error puzzle game rather than an educational simulation of valence electron sharing. By pausing gameplay at stage transitions (StageTransitionModal.jsx) and domain clearances (DebriefingModal.jsx), the system forces cognitive consolidation—explicitly linking the visual reaction to valence electron pairing, electronegativity differences, and Lewis structures before the learner faces more complex stoichiometry.

7. **Expected improvement**
    - **Cognitive Mastery:** Raises post-test chemical bonding concept discrimination accuracy from baseline 68.6% to >= 80.0%, achieving normalized learning gain g >= 0.50 (high gain on Hake's scale).
    - **Pedagogical Alignment:** Bridges curriculum standards (DepEd Grade 9 Chemical Bonding Competency), resolving teacher feedback regarding abrupt, unguided level progressions.
    - **Learner Relevance:** Increases Keller's ARCS Relevance construct rating from 4.19 to >= 4.50/5.00 by explicitly explaining real-world utility and chemical theory.
8. **How will the improvement eventually be measured?**
    - **Accuracy rate —** Assessed via the 3-item cognitive post-test probe, specifically measuring correct discrimination between covalent electron sharing vs. ionic electron transfer on Question 24.
    - **Task completion rate —** Tracked via Supabase gameplay telemetry recording the percentage of students who successfully review debriefing screens and progress to subsequent cavern stages (>= 95%).
    - **User satisfaction —** Evaluated using Keller's ARCS post-validation survey, targeting Relevance and Satisfaction subscale scores >= 4.50/5.00.
9. **Priority**

    🔴 High — Essential to achieving a SMART Objective 2.

    Justification: Directly addresses the project's primary educational purpose. Without milestone debriefing, the system fails to achieve the mandated g >= 0.40 learning gain and 75% curriculum mastery threshold specified in SMART Objective 2.

Refactoring Priority #2: Centralized Diagnostic Pre-Test Cloud Sync

1. **Validation Finding**

    Referencing Finding #2 from Section 3 (Pre-Test Diagnostic Telemetry Fragmentation & Client-Side Isolation).

2. **What needs to be improved or changed?**

    Refactor PreTestModal.jsx to dispatch an asynchronous POST /api/telemetry/pretest request to Spring Boot, persisting baseline scores and serialized choices directly into a centralized Supabase PRETEST_DIAGNOSTIC table, while retaining localStorage as an offline fallback.

3. **What type of refactoring is required?**
    - **Functional requirement —** Introduces FR-2.4 specifying the automated transmission, verification, and persistence of baseline diagnostic scores upon pre-test completion.
    - **System architecture/design —** Establishes a dedicated REST controller endpoint (PreTestTelemetryController.java) and service layer linking client telemetry directly to Supabase cloud persistence.
    - **Data/input/output —** Defines the formal JSON diagnostic telemetry payload (score, total, percentage, answers_json, device_info) and creates the PRETEST_DIAGNOSTIC PostgreSQL cloud schema.
4. **Current MVP approach:**

    Pre-test diagnostic scores were cached exclusively in the client browser's localStorage, vulnerable to device switching or cache clearing.

5. **Proposed refactored approach**

    Upon pre-test completion, the client serializes score, total, percentage, and choices into a JSON payload and dispatches an asynchronous REST call. The server commits the record to Supabase in < 2 seconds. If offline, the payload enqueues locally and auto-flushes upon reconnection.

6. **Why is this change necessary?**

    In the baseline MVP, storing pre-test diagnostic records exclusively within client browser localStorage created severe data fragility. During Phase 1 validation, 40% (12 of 30) of student baseline records were permanently severed due to terminal reassignments in school computer labs, accidental cookie/cache purges, and private browsing sessions. Because Hake's normalized gain formula requires an unbroken pair of pre-test and post-test scores (g = (Post - Pre) / (100 - Pre)), lost baseline records incapacitated automated evaluation and forced time-consuming manual paper reconciliation. Centralizing diagnostic telemetry into a dedicated cloud table guarantees 100% data pairing, survives browser cache resets, and enables instant server-side computation of student learning trajectories.

7. **Expected improvement**
    - **Data Capture & Pairing:** 100% persistent cloud capture of student pre-test baselines in Supabase, completely eliminating orphaned post-test sessions.
    - **Telemetry Reliability:** Telemetry record loss reduced from 40.0% to 0.0%, ensuring tamper-proof longitudinal tracking.
    - **Analytics Automation:** Instantaneous calculation of cohort normalized learning gains (g) and cognitive delta scores within the Teacher Mastery Analytics view.
    - **Asynchronous Performance:** Sub-2.0-second HTTPS write latency without impeding guest onboarding flow or gameplay responsiveness.
8. **How will the improvement eventually be measured?**
    - **Accuracy rate —** Validated by 100% data parity between serialized client-submitted pre-test answer arrays and committed records in the Supabase PRETEST_DIAGNOSTIC table.
    - **Error rate —** Telemetry packet drop rate and unhandled database write

exceptions reduced to 0.0% across all validated sessions.

- **System response time —** Automated verification that end-to-end HTTPS POST round-trip latency remains strictly under 2.0 seconds (< 2000ms) on standard 3G/4G network profiles.
- **Efficiency measure —** Reduction of researcher/evaluator manual data collation and paper test reconciliation time from ~4.5 hours per cohort to 0.0 minutes (100% automated).

9. **Priority:**

   🔴 High — Essential to achieving a SMART objective 2 and objective 4.

   Justification: Mathematically prerequisite to calculating learning gains for SMART objective 2, and directly fulfills SMART objective 4 requirement for automated cloud session telemetry logging within 2 seconds.

Refactoring Priority #3: Realtime Heartbeat and Disconnect Engine

1. **Validation Finding**

   Reference Finding #3 from Section 3 (Multiplayer State Vulnerability to Peer Network Dropouts).

2. **What needs to be improved or changed?**

   Implement a background WebSocket heartbeat mechanism emitting pings every 3 seconds during multiplayer matches. An 8-second silence pauses the match timer and renders a reconnection overlay; if reconnection fails after 15 seconds, the engine awards an automatic forfeit victory.

3. **What type of refactoring is required?**
   - **Functional requirement —** Introduces FR-3.4 defining real-time ping/pong emission, disconnect detection triggers, countdown timer freezing, and automated forfeit victory assignment.
   - **System process/workflow —** Modifies the match lifecycle state machine by adding DISCONNECTED_GRACE and FORFEIT_RESOLVED states between active competition and room termination.
   - **System architecture/design —** Implements an active bidirectional heartbeat presence loop over Supabase Realtime WebSocket channels with server-synchronized timestamps.
   - **Usability —** Renders an informative visual countdown modal ('Opponent Disconnected — Reconnecting in 15s...') ensuring the connected student understands match status instead of experiencing an unexplained freeze.
   - **Security / Reliability —** Eliminates infinite hanging sockets, protects room state integrity, and prevents griefing through deliberate mobile browser exits.
4. **Current MVP approach:**

   Multiplayer relied on standard WebSocket lifecycle events, leaving active players stranded when peer devices experienced silent cellular connection drops.

5. **Proposed refactored approach:**

   The Supabase Realtime channel tracks peer heartbeats. An 8-second absence pauses the match countdown timer and renders an 'Opponent Disconnected' overlay. If the peer reconnects within 15 seconds, the match resumes; if the timeout expires, the game engine logs a FORFEIT_WIN and safely returns the connected player to the lobby.

6. **Why is this change necessary?**

During competitive multiplayer testing in real school environments, mobile network instability (packet loss, cellular handoffs, or browser tab switching) frequently severed peer connections. Because the baseline MVP relied strictly on standard passive WebSocket close events, silent network drops failed to emit formal disconnect notifications. Consequently, the remaining student was left stranded in an indefinite game freeze with a locked countdown timer, unable to finish the match, claim a legitimate win, or exit without manually refreshing the browser—which wiped their match streak and corrupted session analytics. This flaw caused significant student frustration during Phase 2 trials, dragging competitive satisfaction down to 3.82/5.0. Implementing an active heartbeat monitor with an automated forfeit guarantee is essential to ensure that every match concludes cleanly and fairly regardless of network volatility.

7. **Expected improvement**
    - **State Robustness:** 100% elimination of stranded match sessions (0% orphan WebSocket states) under any network dropout scenario.
    - **Graceful Recovery & Fairness:** Programmatic 15-second reconnection grace period allowing temporary drops to resume seamlessly, while guaranteeing an automated forfeit win if reconnection fails.
    - **Player Satisfaction:** Raises multiplayer Speed-to-Compound satisfaction ratings from 3.82 back above >= 4.50/5.00, fulfilling the >= 4.20 threshold of SMART Objective OBJ-04.
    - **Deterministic Resolution:** Match resolution achieved within a deterministic maximum ceiling of 23 seconds (8s silence detection + 15s grace countdown).
8. **How will the improvement eventually be measured?**
    - **Task completion rate —** Percentage of initiated multiplayer matches that resolve into a definitive terminal state (WIN, LOSS, or FORFEIT_WIN) reaching 100.0%.
    - **Error rate —** Frequency of unhandled frozen game states or abandoned rooms requiring manual browser reload reduced to 0.0%.
    - **User satisfaction —** Assessed via the Keller ARCS Satisfaction subscale for Speed-to-Compound mechanics, targeting composite scores >= 4.50/5.00.
    - **System response time —** Heartbeat ping interval maintained at 3.0-second cadence, with silence detection tripping reliably at exactly 8.0 seconds of latency.
9. **Priority:**

    🔴 High — Essential to achieving a SMART objective 4.

    Justification: Directly protects competitive fairness and user retention in the Speed-to-Compound module, ensuring satisfaction ratings achieve the mandated >= 4.20/5.00 benchmark specified in SMART Objective 4.

Section 5 — Refactoring Summary Matrix Ask teams to summarize their findings in one table:

#	Validation Finding	Evidence	SMART Objective / Outcome Affected	Proposed Refactoring	Expected Improvement	Priority
1	Post-synthesis cognitive gap; lack of explicit bond discrimination.	Q24 accuracy lowest (68.6%); teacher feedback on abrupt stage transitions.	OBJ-02 (Learning Gain g >= 0.40 on Chemical Bonding).	FR-1.6 Post-Stage & Post-Domain Educational Debrief (StageTransition Modal & DebriefingModal).	Bond classification accuracy >= 80%; g >= 0.50.	HIGH
2	Pre-test diagnostic telemetry fragmented; stored only in client localStorage.	12 pre-test records lost in Phase 1 due to cache resets; 0 cloud tables in Supabase.	OBJ-02 & OBJ-04 (Cloud session telemetry persistence < 2s).	FR-2.4 Centralized Diagnostic Pre-Test Cloud Sync (POST /api/telemetry/pretest).	100% cloud baseline persistence; real-time gain computation.	HIGH
3	Multiplayer matches freeze indefinitely upon peer cellular network disconnection.	3 of 18 Phase 2 multiplayer trials stuck on frozen timer upon mobile network drop.	OBJ-04 (Satisfaction & Speed-to-Compound >= 4.2/5.0).	FR-3.4 Realtime Heartbeat & Disconnect Engine (3s ping, 8s pause, 15s forfeit).	0% stranded match states; 100% clean forfeit resolution; satisfaction >= 4.50.	HIGH

Section 6 — Items That Are NOT Refactoring Priorities

Item	Identified?	Action
Bugs/errors (audio sound clip overlap on fast drag, button hover glitch)	Yes	Fix during implementation/testing
Minor UI/UX issues (color contrast on element tile symbols, tooltip font size)	Yes	Improve as appropriate
Missing original requirement	No	Implement/complete (All original MVP requirements were delivered)
Missing essential functional requirement	Yes	Assess and incorporate if justified (Incorporated: FR-1.6, FR-2.4, FR-3.4)
Missing non-functional requirement	Yes	Assess and incorporate if justified (Incorporated: NFR-USA-01 48px Touch Targets)
New feature unrelated to SMART objectives (3D periodic table viewer, avatar skins)	Yes	Generally do not prioritize (Rejected as feature creep)
Improvement to measurable outcome (learning gain g, cloud sync latency)	Yes	Potential refactoring priority (Prioritized: FR-1.6 & FR-2.4)
Change in system approach/design (centralized telemetry, heartbeat presence loop)	Yes	Potential refactoring priority (Prioritized: FR-2.4 & FR-3.4)
Improvement in accuracy/efficiency/effectiveness/performance (touch target bounds)	Yes	Potential refactoring priority (Prioritized: NFR-USA-01 & FR-3.4)

Important Question: If your team identified NO significant requirements or design improvements, explain why:

Response: Not applicable — Our team identified three significant, evidence-based refactoring priorities that directly improve our system's SMART objectives.

Section 7 — Final Team Reflection

1. **What is the single most important thing your team learned from the MVP validation?**

   The single most vital insight gained is that gamification elements (interactive dragging, animated barrier destruction, and sound effects) are highly effective at capturing student attention and engagement, but gamification alone does not guarantee conceptual chemistry understanding. Without immediate, structured pedagogical debriefing that explicitly connects the game reaction to the underlying scientific principles (valence electrons, electron sharing vs. transfer, and Lewis dot structures), students will engage through intuitive trial-and-error rather than constructing durable mental models of chemical bonding.

2. **What is the most important change your team should make based on this learning?**

   The most critical change is elevating the post-synthesis transition from a passive victory scorecard into a curriculum-aligned, interactive Post-Stage and Post-Domain Educational Debrief (FR-1.6), backed by centralized pre-test cloud synchronization (FR-2.4). This ensures that every successful reaction milestone immediately delivers cognitive reinforcement that directly drives measurable learning gains.

3. **How will this change help your project achieve its SMART objectives more effectively?**

   This change directly fulfills SMART Objective 2 by targeting the specific cognitive gap identified during validation (raising chemical bond classification accuracy from 68.6% to >= 80% and achieving normalized learning gains g >= 0.50). Furthermore, centralizing pre-test telemetry ensures that every student's learning trajectory is persistently logged in Supabase in under 2 seconds (fulfilling OBJ-04) without risk of data loss from browser cache resets.

4. **What requirements and/or design documents will need to be updated as a result?**
   - **SRS:** Incorporate FR-1.6 (Educational Debrief), FR-2.4 (Pre-Test Cloud Sync), FR-3.4 (Heartbeat Disconnect Engine), NFR-USA-01 (48px Touch Targets), and the complete Requirements Traceability Matrix (RTM).
   - **SDD:** Incorporate the updated Three-Tier Architecture, StageTransitionModal.jsx and DebriefingModal.jsx component specifications, PRETEST_DIAGNOSTIC Supabase entity schema, and HeartbeatTimeoutHandler sequence flows.
   - **SPMP:** Update sprint milestones to reflect Post-MVP Refactoring (Sprint 3), internal QA verification, and final deployment timelines.
   - **Other: STD (Software Test Documentation):** Add test cases TC-M1-06 (Debrief modal verification), TC-M2-04 (Pre-test cloud sync latency < 2s), and TC-M3-04 (Multiplayer disconnect timeout and forfeit resolution).

Final Declaration

We certify that the findings and proposed refactoring priorities submitted in this form are based on actual MVP validation feedback gathered from our identified stakeholders/users (65 secondary students and 4 licensed science educators). We understand that the proposed refactoring must be justified in relation to our project's SMART objectives and measurable outcomes.

Team Lead: **Danielle Maxine P. Ocampo**
Date: October 17, 2026